



**SERANGOON JUNIOR COLLEGE**  
**General Certificate of Education Advanced Level**  
**Higher 2**

CANDIDATE  
NAME

CLASS

**CHEMISTRY**  
**JC2 Preliminary Examination**  
**Paper 4 Practical**

**9729/04**  
**24 August 2018**  
**2 hr 30 min**

Candidates answer on the Question Paper

**READ THESE INSTRUCTIONS FIRST**

Write your name and class on all the work you hand in.  
Give details of the practical shift and laboratory in the boxes provided.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams, graphs or rough work.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.  
You may lose marks if you do not show your working or if you do not use appropriate units.  
Qualitative Analysis Notes are printed on pages **\_\_\_ & \_\_\_**.

|                   |
|-------------------|
| <b>Shift</b>      |
|                   |
| <b>Laboratory</b> |
|                   |

At the end of the examination, fasten all your work securely together.  
The number of marks is given in the brackets [ ] at the end of each question or part question.

| For Examiner's Use |      |
|--------------------|------|
| 1                  | /13  |
| 2                  | /29  |
| 3                  | /13  |
| TOTAL              | / 55 |

All calculations final answers to 3 or 4 sf (unless answer is exact) and must have units – overall minus 1 except question (f)(i).  
Markers to indicate on front page and tutors to minus from final marks.

Answer **all** the questions in the spaces provided.

# 1 Determination of a value for the solubility product, $K_{sp}$ , of calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$

Calcium iodate is used in the manufacture of disinfectants, antiseptics, and deodorants. Its solubility in water is low. When calcium iodate(V) is mixed with water, the following equilibrium is established.

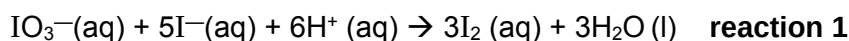


You are to perform an experiment to determine the solubility product,  $K_{sp}$ , of calcium iodate(V).

You will first prepare a saturated solution of calcium iodate(V) by mixing specific volumes of potassium iodate(V),  $\text{KIO}_3$  and calcium nitrate,  $\text{Ca}(\text{NO}_3)_2$ .

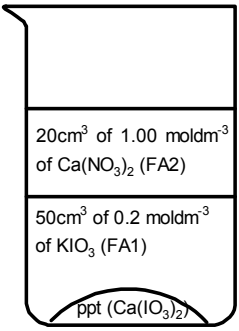
The mixture is then filtered after leaving to stand for some time. The filtrate can then be analysed to determine the amount of iodate(V) ions left as follows:

- Excess potassium iodide, KI, is added to an acidified solution of the filtrate, liberating iodine.



- The liberated iodine is then titrated with a standard solution of sodium thiosulfate.



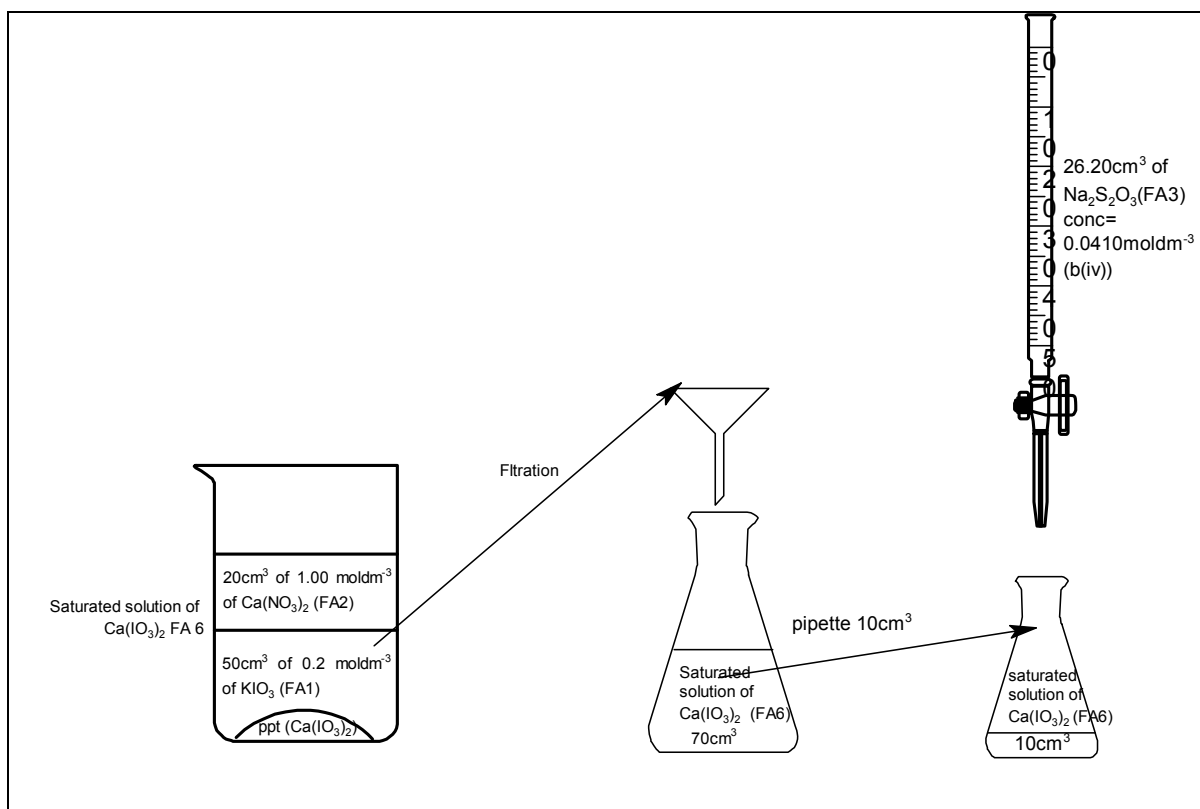
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>FA 1</b> 0.200 mol dm <sup>-3</sup> potassium iodate(V), $\text{KIO}_3$<br><b>FA 2</b> 1.00 mol dm <sup>-3</sup> calcium nitrate, $\text{Ca}(\text{NO}_3)_2$<br><b>FA 3</b> aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$<br><b>FA 4</b> aqueous solution of potassium iodide, KI<br><b>FA 5</b> dilute hydrochloric acid, HCl<br>Starch indicator                                                                                                                                                                                          |
| (a) | <p><b>Preparing the reaction mixture</b></p> <ol style="list-style-type: none"> <li>Transfer 50 cm<sup>3</sup> of <b>FA 1</b> using a measuring cylinder to the beaker labelled <b>reaction mixture</b>.</li> <li>Using another measuring cylinder, transfer 20 cm<sup>3</sup> of <b>FA 2</b> into the same beaker.</li> <li>A precipitate will form. Stir the mixture thoroughly and leave it to stand for several minutes to allow equilibrium to be reached.</li> </ol>  |
|     | <b>While waiting, follow the instructions given in part (b).</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|     |     |                                                                                                                                                                                                                                           |
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|     |     |                                                                                                                                                                                                                                           |
| (b) | (i) | <b>Dilution of FA 1</b><br><br>The concentration of <b>FA 1</b> provided is too high. You will first prepare a diluted solution of <b>FA 1</b> of known concentration and use it to standardise the sodium thiosulfate solution provided. |
|     |     | 1. Using a burette, transfer 10.00 cm <sup>3</sup> of <b>FA 1</b> into a 250 cm <sup>3</sup> graduated flask, labelled <b>diluted FA 1</b> .                                                                                              |
|     |     | 2. Make up to the mark with deionised water and mix thoroughly.                                                                                                                                                                           |

|  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | (ii) | <b>Standardisation of FA 3</b><br>1. Fill a burette with <b>FA 3</b> .<br>2. Pipette 25.0 cm <sup>3</sup> of <b>diluted FA 1</b> into a conical flask.<br>3. Using a measuring cylinder, add about 10 cm <sup>3</sup> of <b>FA 4</b> to the flask.<br>4. Using another measuring cylinder, add about 2 cm <sup>3</sup> of <b>FA 5</b> to the flask.<br>5. Add <b>FA 3</b> from the burette into the flask until a pale yellow colour is obtained.<br>6. Add about 5 drops of starch indicator into the flask. Continue adding <b>FA 3</b> until the blue-black colour just disappears.<br>7. Perform sufficient titrations to obtain accurate results for the end-point. Rinse the conical flask between each titration. |
|  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                              |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
|----------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------|-------|------------------------------------------|------|------|----------------------------------------------|-------|-------|--|---|---|
|                                              |       | Record your titration results in the space below. Make certain that your recorded results show the precision of your working. [5]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
|                                              |       | <table border="1"> <tr> <td>Final burette readings / cm<sup>3</sup></td><td>29.25</td><td>29.35</td></tr> <tr> <td>Initial burette reading /cm<sup>3</sup></td><td>0.00</td><td>0.00</td></tr> <tr> <td>Volume of <b>FA 3</b> used / cm<sup>3</sup></td><td>29.25</td><td>29.35</td></tr> <tr> <td></td><td>√</td><td>√</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                     | Final burette readings / cm <sup>3</sup> | 29.25 | 29.35 | Initial burette reading /cm <sup>3</sup> | 0.00 | 0.00 | Volume of <b>FA 3</b> used / cm <sup>3</sup> | 29.25 | 29.35 |  | √ | √ |
| Final burette readings / cm <sup>3</sup>     | 29.25 | 29.35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
| Initial burette reading /cm <sup>3</sup>     | 0.00  | 0.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
| Volume of <b>FA 3</b> used / cm <sup>3</sup> | 29.25 | 29.35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
|                                              | √     | √                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
|                                              | (iii) | From your titration results, obtain a suitable volume of <b>FA 3</b> to be used in your calculations. Show clearly how you obtained this volume. [1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
|                                              |       | Volume of <b>FA 3</b> used = (29.25 + 29.35) / 2 = 29.30 cm <sup>3</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
|                                              | (iv)  | Calculate the concentration of S <sub>2</sub> O <sub>3</sub> <sup>2-</sup> ions in the <b>FA 3</b> solution. [1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |
|                                              |       | <p>Amount of KIO<sub>3</sub> in 250 cm<sup>3</sup> of <b>diluted FA 1</b><br/>         = Amount of KIO<sub>3</sub> in 10 cm<sup>3</sup> of <b>concentrated FA 1</b><br/> <math>= 0.2 \times \frac{10}{1000} = 0.002 \text{ mol}</math></p> <p>Amount of KIO<sub>3</sub> in 25 cm<sup>3</sup> of <b>diluted FA 1</b> = 0.002/10 = 0.0002 mol</p> <p>IO<sub>3</sub><sup>-</sup> ≡ 3I<sub>2</sub> ≡ 6S<sub>2</sub>O<sub>3</sub><sup>2-</sup></p> <p>Amount of S<sub>2</sub>O<sub>3</sub><sup>2-</sup> in 29.30 cm<sup>3</sup> of <b>FA 3</b> = 0.0002 x 6 = 0.0012 mol</p> <p>[S<sub>2</sub>O<sub>3</sub><sup>2-</sup>] of <b>FA3</b> = 0.0012 / (<math>\frac{29.30}{1000}</math>) = <b><u>0.0410 mol dm<sup>-3</sup></u></b></p> |                                          |       |       |                                          |      |      |                                              |       |       |  |   |   |

|     |     |                                                                                                                                                                                                            |
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|     |     | <b>Analysing the filtrate</b>                                                                                                                                                                              |
| (c) | (i) | 1. Filter the reaction mixture through a <b>dry</b> filter paper into a <b>dry</b> conical flask, labelled <b>FA 6</b> . This is the filtrate, <b>FA 6</b> . Do not wash the white precipitate with water. |
|     |     | 2. Pipette 10.0 cm <sup>3</sup> of <b>FA 6</b> into a conical flask.                                                                                                                                       |
|     |     | 3. Using a measuring cylinder, add about 10 cm <sup>3</sup> of <b>FA 4</b> to the flask.                                                                                                                   |
|     |     | 4. Using another measuring cylinder, add about 2 cm <sup>3</sup> of <b>FA 5</b> to the flask.                                                                                                              |
|     |     | 5. Add <b>FA 3</b> from the burette into the flask until a pale yellow solution is obtained.                                                                                                               |
|     |     | 6. Add about 5 drops of starch indicator and continue adding <b>FA 3</b> until the blue-black colour just disappears.                                                                                      |
|     |     | 7. Perform sufficient titrations to obtain accurate results for the end-point. Rinse the conical flask between each titration.                                                                             |



Record your titration results in the space below. Make certain that your recorded results show the precision of your working.

|                                              |       |       |
|----------------------------------------------|-------|-------|
| Final burette readings / cm <sup>3</sup>     | 26.20 | 26.20 |
| Initial burette reading/cm <sup>3</sup>      | 0.00  | 0.00  |
| Volume of <b>FA 3</b> used / cm <sup>3</sup> | 26.20 | 26.20 |
|                                              | ✓     | ✓     |

(ii) From your titration results, obtain a suitable volume of **FA 3** to be used in your calculations. Show clearly how you obtained this volume.

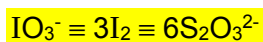
$$\text{Volume of FA 3 used} = (26.20 + 26.20) / 2 = 26.20 \text{ cm}^3$$

#### Calculations

(d) (i) Calculate the amount of S<sub>2</sub>O<sub>3</sub><sup>2-</sup> ions present in the volume of **FA 3** obtained in (c)(ii).

$$\text{Amount of S}_2\text{O}_3^{2-} = 0.0410 \times \left( \frac{26.20}{1000} \right) = \underline{0.00107 \text{ mol}}$$

(ii) Calculate the amount of IO<sub>3</sub><sup>-</sup> ions present in 10.0 cm<sup>3</sup> of the filtrate, **FA 6**. [1]



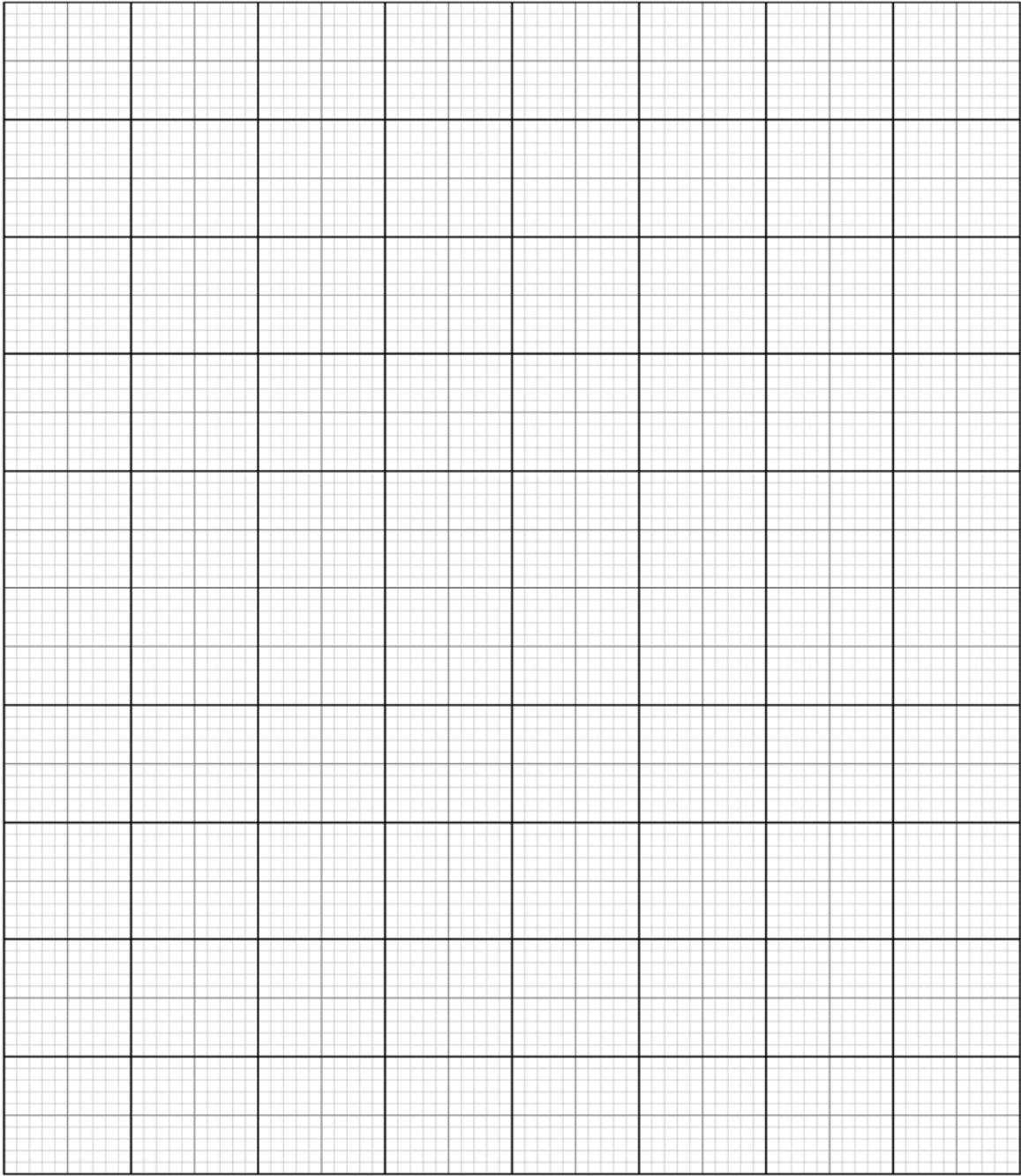
$$\text{Amount of IO}_3^- \text{ in } 10.0 \text{ cm}^3 \text{ of FA 6} = 0.0010742 / 6 = \underline{1.79 \times 10^{-4} \text{ mol}}$$

|     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | (iii) | Hence, calculate the total amount of $\text{IO}_3^-$ ions present in the filtrate, <b>FA 6</b> . [1]<br><br>Total volume of mixture = $50 + 20 = 70 \text{ cm}^3$<br><br>Amount of $\text{IO}_3^-$ in the filtrate ( $70 \text{ cm}^3$ of <b>FA 6</b> )<br>= $1.79 \times 10^{-4} \times (70/10) = \underline{1.25 \times 10^{-3} \text{ mol}}$                                                                                                                                                                       |
| (e) | (i)   | Calculate the initial amount of $\text{IO}_3^-$ ions and $\text{Ca}^{2+}$ ions present in the reaction mixture prepared in (a). [1]<br><br>Initial amount of $\text{IO}_3^- = 50/1000 \times 0.2 = \underline{0.0100 \text{ mol}}$<br><br>Initial amount of $\text{Ca}^{2+} = 20/1000 \times 1.0 = \underline{0.0200 \text{ mol}}$                                                                                                                                                                                    |
|     | (ii)  | Calculate the amount of $\text{IO}_3^-$ ions precipitated as $\text{Ca}(\text{IO}_3)_2$ . [1]<br><br>Amount of $\text{IO}_3^-$ precipitated as $\text{Ca}(\text{IO}_3)_2 = \text{Initial amount} - \text{amount in filtrate}$<br>= $0.0100 - (1.25 \times 10^{-3}) = 0.00875 \text{ mol}$                                                                                                                                                                                                                             |
|     | (iii) | Hence, calculate the amount of $\text{Ca}^{2+}$ ions left in <b>FA 6</b> . [1]<br>$\text{Ca}^{2+} \equiv 2\text{IO}_3^-$<br><br>Amount of $\text{Ca}^{2+}$ precipitated = $0.00875 / 2 = 0.004375 \text{ mol}$<br><br>Amount of $\text{Ca}^{2+}$ ions left in <b>FA 6</b> = Initial amount – amount in precipitate<br>= $0.02 - 0.004375 = \underline{0.0156 \text{ mol}}$                                                                                                                                            |
| (f) | (i)   | Use your answer in parts (d)(iii) and (e)(iii) to calculate a value for the solubility product, $K_{sp}$ , of calcium iodate(V). Include units in your answer. [2]<br><br>$\text{Eqm } [\text{Ca}^{2+}] = \frac{0.015625}{0.07} = 0.223 \text{ mol dm}^{-3}$<br><br>$\text{Eqm } [\text{IO}_3^-] = \frac{1.25 \times 10^{-3}}{0.07} = 0.01786 \text{ mol dm}^{-3}$<br><br>$K_{sp} = [\text{Ca}^{2+}][\text{IO}_3^-]^2 = (0.223)(0.01786)^2 = \underline{7.11 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}}$           |
| (g) |       | Another student performed this experiment and obtained a value for the solubility product, $K_{sp}$ , of $3.45 \times 10^{-5}$ . A literature value for this solubility product is $6.71 \times 10^{-6}$ at $20^\circ\text{C}$ .<br><br>You should assume that apparatus of the same precision was used in each case.<br><br>State a possible reason for the higher value of $K_{sp}$ obtained by the student and suggest an improvement which might allow a value closer to the literature value to be obtained. [1] |
|     |       | <b>Explanation:</b><br><br>The student did not carry out the experiment at $20^\circ\text{C}$ . Hence, the equilibrium position shifted to the right.<br><br>OR                                                                                                                                                                                                                                                                                                                                                       |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>When the student carried out the experiment, not all the ppt has formed / eqm has not been reached. Hence, the <math>[\text{Ca}^{2+}]</math> and <math>[\text{IO}_3^-]</math> is too high.</p> <p><b>Improvement:</b></p> <p>The student should <u>carry out the experiment at 20°C</u> by letting the mixture equilibrate in a thermostatically <u>controlled</u> water bath <u>maintained</u> at 20°C for a <u>considerable amount of time</u> before doing the filtration.</p> <p>OR</p> <p>The student should leave the reaction mixture for a <u>considerable time</u> before doing the filtration to <u>allow</u> the ppt to form / <u>equilibrium to be establish</u>.</p> |
|  | [Total: 15 ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | <p>You are to determine the enthalpy change of reaction, <math>\Delta H</math>, for the reaction shown below.</p> $\text{Cu(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{H}_2(\text{g})$ <p>As copper is an unreactive metal it does not react with dilute acids. You will need to find the enthalpy change of reaction for two reactions that do occur. The equations for these two reactions are below.</p> $\text{Mg(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{H}_2(\text{g}) \quad \textbf{Reaction 1}$ $\text{Mg(s)} + \text{CuSO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{Cu(s)} \quad \textbf{Reaction 2}$ <p>You will conduct experiments to find the enthalpy changes for each of <b>Reaction 1</b> and <b>Reaction 2</b> and use these values to calculate the enthalpy change for the reaction of copper with sulfuric acid.</p> |
|     | <p><b>Determining the enthalpy change for Reaction 1</b></p> $\text{Mg(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{H}_2(\text{g}) \quad \textbf{Reaction 1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| (a) | <p><b>Method</b></p> <p><b>FA 8</b> is 1.00 mol dm<sup>-3</sup> sulfuric acid, H<sub>2</sub>SO<sub>4</sub>.<br/> <b>FA 7</b> is magnesium powder, Mg.</p> <p>Read through the method before you start any practical work and prepare a suitable table for your results.</p> <ol style="list-style-type: none"> <li>1. Weigh the bottle containing <b>FA 7</b>. Record the mass.</li> <li>2. Support the Styrofoam cup in the 250 cm<sup>3</sup> beaker.</li> <li>3. Use the measuring cylinder to transfer 25 cm<sup>3</sup> of <b>FA 8</b> into the Styrofoam cup.</li> <li>4. Measure the temperature of <b>FA 8</b> in the cup and start the stop clock. Record this temperature as being the temperature at time = 0.</li> <li>5. Measure, and record, the temperature of this <b>FA 8</b> every half minute for 2</li> </ol>                                                                                                              |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>minutes.</p> <ol style="list-style-type: none"><li>6. At time = 2½ minutes add the <b>FA 7</b> to the acid and stir carefully to reduce acid spray.</li><li>7. Measure the temperature of the mixture in the cup at time = 3 minutes and then every half minute up to time = 7 minutes.</li><li>8. Continue stirring occasionally throughout this time.</li><li>9. Weigh the bottle that had contained <b>FA 7</b>. Record the mass.</li><li>10. Calculate and record the mass of <b>FA 7</b> added to the sulfuric acid.</li><li>11. Rinse the cup with water and shake to dry.</li></ol> <p>[6]</p> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                     |                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(b) (i)</b>                                                                      | On the grid below plot a graph of temperature (y-axis) against time (x-axis).                                                                                                                                                                                                    |
|  |                                                                                                                                                                                                                                                                                  |
| <div style="text-align: right;">[2]</div>                                           |                                                                                                                                                                                                                                                                                  |
| <b>(ii)</b>                                                                         | Complete the graph by drawing two, straight lines of best fit <ul style="list-style-type: none"> <li>• One to show the temperature up to time = 2 ½ minutes</li> <li>• One to show the temperature after time = 2 ½ minutes</li> </ul> <div style="text-align: right;">[1]</div> |
| <b>(iii)</b>                                                                        | From your graph, use the two straight lines of best fit to calculate the change in temperature at time = 2 ½ minutes.                                                                                                                                                            |
|                                                                                     |                                                                                                                                                                                                                                                                                  |

|  |  |                                     |
|--|--|-------------------------------------|
|  |  | Temperature change = .....°C<br>[1] |
|--|--|-------------------------------------|

|                         |                                                                                                                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(c) Calculations</b> |                                                                                                                                                                                                                                                                                                        |
| <b>(i)</b>              | In the reaction in <b>(a)</b> , the sulfuric acid was in excess. Without carrying out any additional tests, what observation could you have made during your experiment to confirm this?                                                                                                               |
|                         | <p>All the magnesium / solid dissolved / disappeared or all solid / Mg has gone / been used up or no solid / Mg left.</p> <p><b>Not accepted: no more effervescence.</b></p>                                                                                                                           |
| <b>(ii)</b>             | Calculate the energy change that occurred during the reaction in <b>(a)</b> .<br>[Assume that 4.2 J is needed to raise the temperature of 1.0 cm <sup>3</sup> of solution by 1.0 °C.]                                                                                                                  |
|                         | <p>Correct use of <math>Q = mc\Delta T</math></p> <p style="text-align: right;">[1]</p>                                                                                                                                                                                                                |
| <b>(iii)</b>            | Use your answer to <b>(ii)</b> to calculate the enthalpy change, in kJ mol <sup>-1</sup> , for the reaction between sulfuric acid and magnesium.<br>[Ar : Mg, 24.3]                                                                                                                                    |
|                         | $\text{Mg(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)} \quad \text{Reaction 1}$                                                                                                                                                                  |
|                         | <p>number of moles of magnesium = <math>\frac{\text{mass measured}}{24.3}</math> mol</p> <p><math display="block">\Delta H = - \frac{\text{Ans in (ii)} / 1000}{\text{no. of mol of Mg}}</math></p> <p style="text-align: right;">Enthalpy change of Reaction 1 = .....kJ mol<sup>-1</sup><br/>[1]</p> |

|                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>(d) Planning: Determining the enthalpy change for Reaction 2</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                     | <p>The enthalpy change of reaction for <b>Reaction 2</b> can be determine using a <b>non-graphical approach</b>.</p> $\text{Mg(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{Cu(s)} \quad \text{Reaction 2}$                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>(i)</b>                                                          | <p>Plan an investigation to determine mean temperature rise, mean mass of magnesium used and the enthalpy change of reaction for <b>Reaction 2</b>.</p> <p>You are provided with</p> <ul style="list-style-type: none"> <li>• <b>FA 11</b>, 1.00 mol dm<sup>-3</sup> copper(II) sulfate, CuSO<sub>4</sub></li> <li>• <b>FA 9</b> and <b>FA 10</b> magnesium powder samples in weighing bottles</li> <li>• the equipment normally found in a school or college laboratory.</li> </ul> <p>You are to take precaution to ensure that the copper(II) sulfate was in excess in these reactions and <b>not more than 30 cm<sup>3</sup></b> is used for each experiment.</p> |

|                                           |                         | <p>In your plan you should include brief details of</p> <ul style="list-style-type: none"> <li>the volume of <math>\text{CuSO}_4</math> you plan to use (you may assume the mass of <math>\text{Mg(s)}</math> used to be the same as in reaction 1),</li> <li>the apparatus that you would use,</li> <li>the procedure that you would follow and the measurements that you would take,</li> <li>a suitable table for your results.</li> </ul> <p style="text-align: right;">[6]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                         |                         |                                           |  |  |                                  |  |  |                                |  |  |                     |  |  |                                        |  |  |
|-------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------|-------------------------|-------------------------------------------|--|--|----------------------------------|--|--|--------------------------------|--|--|---------------------|--|--|----------------------------------------|--|--|
|                                           |                         | <p><b>Suggested Answer:</b></p> <ol style="list-style-type: none"> <li>(1) Weigh the weighing bottle containing <b>FA 9</b>. Record the mass.</li> <li>(2) Support the plastic cup in the <math>250\text{ cm}^3</math> beaker.</li> <li>(3) Use the measuring cylinder to transfer <math>25\text{ cm}^3</math> of <b>FA 11</b> into the plastic cup.</li> <li>(4) Measure the temperature of <b>FA 11</b> in the plastic cup and record the temperature.</li> <li>(5) Add the <b>FA 9</b> to the <b>FA 11</b> in the cup and stir the mixture constantly.</li> <li>(6) Measure and record the maximum temperature, reached during the reaction.</li> <li>(7) Calculate and record the maximum temperature change that occurred during the reaction between <b>FA 9</b> and <b>FA 11</b>.</li> <li>(8) Weigh the stoppered tube that had contained <b>FA 9</b>. Record the mass.</li> <li>(9) Calculate and record the mass of <b>FA 9</b> added to the copper(II) sulfate.</li> <li>(10) Empty the contents of the plastic cup into the <math>100\text{ cm}^3</math> beaker labelled <b>waste</b>.</li> <li>(11) Rinse the plastic cup and shake to dry or dry with the cleaning paper provided.</li> <li>(12) Repeat step (1) to (9) experiment using <b>FA 10</b> in place of <b>FA 9</b>.</li> </ol> <table border="1"> <thead> <tr> <th></th><th>Expt with FA 3 and FA 4</th><th>Expt with FA 3 and FA 5</th></tr> </thead> <tbody> <tr> <td>Volume of <math>\text{CuSO}_4</math> / <math>\text{cm}^3</math></td><td></td><td></td></tr> <tr> <td>Initial mass of tube with Mg / g</td><td></td><td></td></tr> <tr> <td>Final mass of tube with Mg / g</td><td></td><td></td></tr> <tr> <td>Mass of Mg used / g</td><td></td><td></td></tr> <tr> <td>Initial temperature / <math>^\circ\text{C}</math></td><td></td><td></td></tr> </tbody> </table> |  | Expt with FA 3 and FA 4 | Expt with FA 3 and FA 5 | Volume of $\text{CuSO}_4$ / $\text{cm}^3$ |  |  | Initial mass of tube with Mg / g |  |  | Final mass of tube with Mg / g |  |  | Mass of Mg used / g |  |  | Initial temperature / $^\circ\text{C}$ |  |  |
|                                           | Expt with FA 3 and FA 4 | Expt with FA 3 and FA 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                         |                         |                                           |  |  |                                  |  |  |                                |  |  |                     |  |  |                                        |  |  |
| Volume of $\text{CuSO}_4$ / $\text{cm}^3$ |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                         |                         |                                           |  |  |                                  |  |  |                                |  |  |                     |  |  |                                        |  |  |
| Initial mass of tube with Mg / g          |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                         |                         |                                           |  |  |                                  |  |  |                                |  |  |                     |  |  |                                        |  |  |
| Final mass of tube with Mg / g            |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                         |                         |                                           |  |  |                                  |  |  |                                |  |  |                     |  |  |                                        |  |  |
| Mass of Mg used / g                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                         |                         |                                           |  |  |                                  |  |  |                                |  |  |                     |  |  |                                        |  |  |
| Initial temperature / $^\circ\text{C}$    |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                         |                         |                                           |  |  |                                  |  |  |                                |  |  |                     |  |  |                                        |  |  |

|                                                                                                                                   |      | <table><tr><td>Final temperature/ °C</td><td></td><td></td></tr><tr><td>Change in temperature / °C</td><td></td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Final temperature/ °C |      |                                                                                          | Change in temperature / °C |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
|-----------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------|------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------|---|-----------------------------------|---|---------------------------------------------------------------------------------------------------------|---|------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------|---|
| Final temperature/ °C                                                                                                             |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Change in temperature / °C                                                                                                        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
|                                                                                                                                   |      | <table><tr><th>Indicative material</th><th>Mark</th></tr><tr><td>Relevant working for proposed volume of CuSO<sub>4</sub> to ensure that it is in excess</td><td>1</td></tr><tr><td>Proper use of apparatus</td><td>1</td></tr><tr><td>Sound process in taking mass</td><td>1</td></tr><tr><td>Rinse cup and dry before conducting second expt or use a new cup</td><td>1</td></tr><tr><td>Repeating the experiment using FA5</td><td>1</td></tr><tr><td>Table with at 8 readings<br/>· 4 × balance readings<br/>· 2 × initial temp<br/>· 2 × highest / max temp<br/>with unambiguous headings</td><td>1</td></tr></table> <p>If student copied the procedure in front, but instead of drawing graph, just take max T from table, penalise under sound process as they may miss the actual T max.</p> | Indicative material   | Mark | Relevant working for proposed volume of CuSO <sub>4</sub> to ensure that it is in excess | 1                          | Proper use of apparatus                                            | 1 | Sound process in taking mass      | 1 | Rinse cup and dry before conducting second expt or use a new cup                                        | 1 | Repeating the experiment using FA5 | 1 | Table with at 8 readings<br>· 4 × balance readings<br>· 2 × initial temp<br>· 2 × highest / max temp<br>with unambiguous headings | 1 |
| Indicative material                                                                                                               | Mark |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Relevant working for proposed volume of CuSO <sub>4</sub> to ensure that it is in excess                                          | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Proper use of apparatus                                                                                                           | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Sound process in taking mass                                                                                                      | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Rinse cup and dry before conducting second expt or use a new cup                                                                  | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Repeating the experiment using FA5                                                                                                | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Table with at 8 readings<br>· 4 × balance readings<br>· 2 × initial temp<br>· 2 × highest / max temp<br>with unambiguous headings | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
|                                                                                                                                   | (ii) | <p><b>Perform</b> the experiment you have planned in <b>(d)(i)</b>, clearly calculating the</p> <ul style="list-style-type: none"><li>• mean temperature change and</li><li>• mean mass of magnesium.</li></ul> <p>Hence, determine the enthalpy change, in kJ mol<sup>-1</sup> for the reaction between magnesium and copper(II) sulfate.</p> <p>[Assume that 4.2 J is needed to raise the temperature of 1.0 cm<sup>3</sup> of solution by 1.0°C]<br/>[A<sub>r</sub> : Mg, 24.3]</p> <p>[3]</p>                                                                                                                                                                                                                                                                                                     |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
|                                                                                                                                   |      | <table><tr><th>Indicative material</th><th>Mark</th></tr><tr><td>Proper recording of expt data (as in 1(a))</td><td>1</td></tr><tr><td>Conducted the experiment and determine the mean temperature change</td><td>1</td></tr><tr><td>Mean mass of magnesium determined</td><td>1</td></tr><tr><td><b>Working</b> to calculate the enthalpy change using mean mas of magnesium and mean temperature change</td><td>1</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                 | Indicative material   | Mark | Proper recording of expt data (as in 1(a))                                               | 1                          | Conducted the experiment and determine the mean temperature change | 1 | Mean mass of magnesium determined | 1 | <b>Working</b> to calculate the enthalpy change using mean mas of magnesium and mean temperature change | 1 |                                    |   |                                                                                                                                   |   |
| Indicative material                                                                                                               | Mark |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Proper recording of expt data (as in 1(a))                                                                                        | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Conducted the experiment and determine the mean temperature change                                                                | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| Mean mass of magnesium determined                                                                                                 | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |
| <b>Working</b> to calculate the enthalpy change using mean mas of magnesium and mean temperature change                           | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |      |                                                                                          |                            |                                                                    |   |                                   |   |                                                                                                         |   |                                    |   |                                                                                                                                   |   |

**Enthalpy change for Reaction 3**

|     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |      | Reaction 3 is shown below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|     |      | $\text{Cu(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{H}_2(\text{g}) \quad \text{Reaction 3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| (e) |      | <p>Use your values for the enthalpy changes for <b>Reactions 1</b> and <b>2</b> to calculate the enthalpy change for <b>Reaction 3</b>.</p> $\text{Mg(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{H}_2(\text{g}) \quad \text{Reaction 1}$ $\text{Mg(s)} + \text{CuSO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{Cu(s)} \quad \text{Reaction 2}$ <p>Show clearly how you obtain your answer.</p> <p>(If you were unable to calculate the enthalpy changes for <b>Reactions 1</b> and <b>2</b>, you should assume that the value for <b>Reaction 1</b> is <math>-444 \text{ kJ mol}^{-1}</math> and that the value for <b>Reaction 2</b> is <math>-504 \text{ kJ mol}^{-1}</math>. Note: these are not the correct values.)</p>       |
|     |      | <p>Method 1 (algebraic method):</p> $\cancel{\text{Mg(s)}} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \cancel{\text{MgSO}_4(\text{aq})} + \text{H}_2(\text{g}) \quad \text{Reaction 1}$ $\cancel{\text{MgSO}_4(\text{aq})} + \text{Cu(s)} \rightarrow \cancel{\text{Mg(s)}} + \text{CuSO}_4(\text{aq}) \quad - \text{Reaction 2}$ <p>Overall: <math>\text{Cu(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{H}_2(\text{g}) \quad \text{Reaction 3}</math></p> <p>Thus, <math>\Delta H_{\text{rxn3}} = \Delta H_{\text{rxn1}} - \Delta H_{\text{rxn2}}</math></p> <p>Method 2 (energy cycle method) is also acceptable.</p> <p style="text-align: right;">Enthalpy change for <b>Reaction 3</b> = ..... <math>\text{kJ mol}^{-1}</math><br/>[2]</p> |
| (f) | (i)  | <p>The method you used to determine the enthalpy change for <b>Reaction 1</b> was more accurate than the method you used to determine the enthalpy change for <b>Reaction 2</b>. Suggest <b>two</b> reasons why the method used for <b>Reaction 2</b> was less accurate. Explain your answers.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|     |      | <p>Any 2 accepted:</p> <ul style="list-style-type: none"> <li>• Lower <math>\Delta H</math> for Reaction 2 and so higher % error</li> <li>• No correction made for loss of heat on cooling using temperature-correction graph, thus heat loss to surroundings for Reaction 2 not taken into account</li> <li>• Not all Mg reacted / reaction does not go to completion in Reaction 2 (so not all energy released)</li> <li>• Reaction 2 involving Mg takes place more slowly, so more heat was lost to surroundings</li> </ul>                                                                                                                                                                                                                                                              |
|     | (ii) | <p>A student suggested that the accuracy of the method used for <b>Reaction 2</b> could be improved by using a larger volume of copper(II) sulfate. Is this a correct suggestion? Give a reason for your answer.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|  |  |                                                                                                                                                                                                                                                                                                                                                                                      |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <p>No, since (larger volume of solution means) smaller <math>\Delta T</math>, hence higher percentage error.</p> <p>OR</p> <p>Yes, since there would be a smaller T rise so less heat would be lost.<br/>Yes, since larger volume is used, hence smaller percentage error</p> <p><b>Not accepted: did not indicate whether the percentage error is for temperature or volume</b></p> |
|  |  | [Total: 29]                                                                                                                                                                                                                                                                                                                                                                          |

| 3                                                                                                                                                                | <p><b>FA 12</b> contains two cations and two anions from those listed on pages <b>18 and 19</b>.</p> <p>In all tests, the reagent should be added gradually until no further change is observed, with shaking after each addition.<br/>Record your observations in the spaces provided.<br/>Your answers should include</p> <ul style="list-style-type: none"><li>• details of colour changes and precipitates formed,</li><li>• the names of gases evolved and details of the test used to identify each one.</li></ul> <p>You should indicate clearly at what stage in a test a change occurs, writing any deductions you made alongside the observations on which they are based.</p> <p>Marks are <b>not</b> given for chemical equations.</p> <p><b>No additional or confirmatory tests for ions present should be attempted.</b></p>                                                                    |                          |                  |                        |                                                                                                                                       |                                                                                                                                                |                          |                                                                                                                                                                  |                                                                                                           |                          |                                                                  |                             |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------|-----------------------------|--|
| <b>Candidates are reminded that definite deductions may be made from tests where there appears to be no reaction.</b>                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |                  |                        |                                                                                                                                       |                                                                                                                                                |                          |                                                                                                                                                                  |                                                                                                           |                          |                                                                  |                             |  |
|                                                                                                                                                                  | <table><tr><th>Test</th><th>Observations [3]</th><th>Possible cation(s) [3]</th></tr><tr><td>(a) Place 3 cm<sup>3</sup> <b>FA 12</b> in a test tube, add aqueous sodium hydroxide, drop by drop, until no further change is seen.</td><td><p>✓ <u>Cream/ Off-white ppt formed</u></p><p>✓ <u>Cream/ Off-white ppt turns brown on standing</u></p><p>(traces of white ppt observed)</p></td><td>✓ <u>Mn<sup>2+</sup></u></td></tr><tr><td>(b) Filter the mixture from (a) and collect the filtrate for later test<br/><br/>Leave the residue in the filter paper and observe it again after several minutes.</td><td><p>✓ <u>Colourless filtrate</u></p><p>✓ <u>Off-white ppt turns brown on standing / brown residue</u></p></td><td>✓ <u>Mn<sup>2+</sup></u></td></tr><tr><td>(c) Extract about 1 cm<sup>3</sup> of the filtrate using a test</td><td>✓ <u>White ppt reformed</u></td><td></td></tr></table> | Test                     | Observations [3] | Possible cation(s) [3] | (a) Place 3 cm <sup>3</sup> <b>FA 12</b> in a test tube, add aqueous sodium hydroxide, drop by drop, until no further change is seen. | <p>✓ <u>Cream/ Off-white ppt formed</u></p> <p>✓ <u>Cream/ Off-white ppt turns brown on standing</u></p> <p>(traces of white ppt observed)</p> | ✓ <u>Mn<sup>2+</sup></u> | (b) Filter the mixture from (a) and collect the filtrate for later test<br><br>Leave the residue in the filter paper and observe it again after several minutes. | <p>✓ <u>Colourless filtrate</u></p> <p>✓ <u>Off-white ppt turns brown on standing / brown residue</u></p> | ✓ <u>Mn<sup>2+</sup></u> | (c) Extract about 1 cm <sup>3</sup> of the filtrate using a test | ✓ <u>White ppt reformed</u> |  |
| Test                                                                                                                                                             | Observations [3]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Possible cation(s) [3]   |                  |                        |                                                                                                                                       |                                                                                                                                                |                          |                                                                                                                                                                  |                                                                                                           |                          |                                                                  |                             |  |
| (a) Place 3 cm <sup>3</sup> <b>FA 12</b> in a test tube, add aqueous sodium hydroxide, drop by drop, until no further change is seen.                            | <p>✓ <u>Cream/ Off-white ppt formed</u></p> <p>✓ <u>Cream/ Off-white ppt turns brown on standing</u></p> <p>(traces of white ppt observed)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ✓ <u>Mn<sup>2+</sup></u> |                  |                        |                                                                                                                                       |                                                                                                                                                |                          |                                                                                                                                                                  |                                                                                                           |                          |                                                                  |                             |  |
| (b) Filter the mixture from (a) and collect the filtrate for later test<br><br>Leave the residue in the filter paper and observe it again after several minutes. | <p>✓ <u>Colourless filtrate</u></p> <p>✓ <u>Off-white ppt turns brown on standing / brown residue</u></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ✓ <u>Mn<sup>2+</sup></u> |                  |                        |                                                                                                                                       |                                                                                                                                                |                          |                                                                                                                                                                  |                                                                                                           |                          |                                                                  |                             |  |
| (c) Extract about 1 cm <sup>3</sup> of the filtrate using a test                                                                                                 | ✓ <u>White ppt reformed</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |                  |                        |                                                                                                                                       |                                                                                                                                                |                          |                                                                                                                                                                  |                                                                                                           |                          |                                                                  |                             |  |

|  |                                                                                                               |                                                                |                                                                                                   |
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|  | pipette into a clean test tube.<br><br>Add dilute nitric acid, drop by drop, until no further change is seen. | <u>✓ White ppt soluble in excess <math>\text{HNO}_3</math></u> | <u>✓ <math>\text{Zn}^{2+}</math> or <math>\text{Al}^{3+}</math> (<math>\text{Pb}^{2+}</math>)</u> |
|--|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|

### Test for anions

Test (e) has been conducted and the observation is recorded.

You are to complete test (f) and devise **two** more tests to confirm the identity of the anions present in **FA 12**.

In the space provided below, clearly state the tests, observations and deductions.

|     | Test                                                                                                                                                                                                                                                                                                                                                    | Observation                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| (e) | Add 1 cm <sup>3</sup> of <b>FA 12</b> into a clean test-tube.<br><br>Add 2 cm <sup>3</sup> of aqueous hydrochloric acid.                                                                                                                                                                                                                                | No effervescence observed<br>No brown gas                                                                   |
| (f) | Add 1 cm <sup>3</sup> of <b>FA 12</b> into a clean test-tube<br><br>Add 3 drops of aqueous silver nitrate.                                                                                                                                                                                                                                              | <u>No ppt</u><br>Solution remains colourless<br>Small amount of white/cream ppt which immediately dissolves |
| (g) | <u>To 1 cm<sup>3</sup> FA 12, add 3 drops of aqueous barium nitrate.</u><br><br><u>Then add 1 cm<sup>3</sup> of aqueous nitric acid/ hydrochloric acid.</u><br><br><u>OR</u><br><br><u>Extract about 1 cm<sup>3</sup> of the filtrate using a dropping pipette/dropper into a clean test tube.</u><br><br><u>Add 3 drops of aqueous barium nitrate.</u> | <u>White ppt</u><br><br><br><br><br><br><br><br><br><br><u>White ppt insoluble</u>                          |
| (h) | <u>To 1 cm<sup>3</sup>/ depth FA 12, add (3 drops X) 2 cm<sup>3</sup>/ equal volume/depth of aqueous NaOH follow by a small piece of Al foil with warming</u>                                                                                                                                                                                           | <u>Effervescence observed</u><br><br><u>Gas evolved turns moist/damp red litmus turns blue</u>              |

Anions present in **FA12**: .....and ..... [1]

$\text{SO}_4^{2-}$  and  $\text{NO}_3^-$  both correct

[Total: 13]

END

**Qualitative Analysis Notes**

[ppt. = precipitate]

**(a) Reactions of aqueous cations**

| <b>cation</b>                                  | <b>reaction with</b>                                                                |                                                                                     |
|------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                | NaOH(aq)                                                                            | NH <sub>3</sub> (aq)                                                                |
| aluminium,<br>Al <sup>3+</sup> (aq)            | white ppt.<br>soluble in excess                                                     | white ppt.<br>insoluble in excess                                                   |
| ammonium,<br>NH <sub>4</sub> <sup>+</sup> (aq) | ammonia produced on heating                                                         | –                                                                                   |
| barium,<br>Ba <sup>2+</sup> (aq)               | no ppt.<br>(if reagents are pure)                                                   | no ppt.                                                                             |
| calcium,<br>Ca <sup>2+</sup> (aq)              | white ppt. with high<br>[Ca <sup>2+</sup> (aq)]                                     | no ppt.                                                                             |
| chromium(III),<br>Cr <sup>3+</sup> (aq)        | grey-green ppt.<br>soluble in excess<br>giving dark green solution                  | grey-green ppt.<br>insoluble in excess                                              |
| copper(II),<br>Cu <sup>2+</sup> (aq),          | pale blue ppt.<br>insoluble in excess                                               | blue ppt.<br>soluble in excess<br>giving dark blue solution                         |
| iron(II),<br>Fe <sup>2+</sup> (aq)             | green ppt., turning brown on<br>contact with air<br>insoluble in excess             | green ppt., turning brown on<br>contact with air<br>insoluble in excess             |
| iron(III),<br>Fe <sup>3+</sup> (aq)            | red-brown ppt.<br>insoluble in excess                                               | red-brown ppt.<br>insoluble in excess                                               |
| magnesium,<br>Mg <sup>2+</sup> (aq)            | white ppt.<br>insoluble in excess                                                   | white ppt.<br>insoluble in excess                                                   |
| manganese(II),<br>Mn <sup>2+</sup> (aq)        | off-white ppt., rapidly turning<br>brown on contact with air<br>insoluble in excess | off-white ppt., rapidly turning<br>brown on contact with air<br>insoluble in excess |
| zinc,<br>Zn <sup>2+</sup> (aq)                 | white ppt.<br>soluble in excess                                                     | white ppt.<br>soluble in excess                                                     |

**(b) Reactions of anions**

| <i>ion</i>                             | <i>reaction</i>                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carbonate, $\text{CO}_3^{2-}$          | $\text{CO}_2$ liberated by dilute acids                                                                                                                                                          |
| chloride, $\text{Cl}^-(\text{aq})$     | gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$ )                                                                                                            |
| bromide, $\text{Br}^-(\text{aq})$      | gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$ )                                                                                             |
| iodide, $\text{I}^-(\text{aq})$        | gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$ )                                                                                                         |
| nitrate, $\text{NO}_3^-(\text{aq})$    | $\text{NH}_3$ liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil                                                                                                                     |
| nitrite, $\text{NO}_2^-(\text{aq})$    | $\text{NH}_3$ liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil;<br>$\text{NO}$ liberated by dilute acids<br>(colourless $\text{NO} \rightarrow$ (pale) brown $\text{NO}_2$ in air) |
| sulfate, $\text{SO}_4^{2-}(\text{aq})$ | gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)                                                                                                      |
| sulfite, $\text{SO}_3^{2-}(\text{aq})$ | $\text{SO}_2$ liberated with dilute acids;<br>gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)                                                                 |

**(c) Tests for gases**

| <i>gas</i>                    | <i>test and test result</i>                                                      |
|-------------------------------|----------------------------------------------------------------------------------|
| ammonia, $\text{NH}_3$        | turns damp red litmus paper blue                                                 |
| carbon dioxide, $\text{CO}_2$ | gives a white ppt. with limewater<br>(ppt. dissolves with excess $\text{CO}_2$ ) |
| chlorine, $\text{Cl}_2$       | bleaches damp litmus paper                                                       |
| hydrogen, $\text{H}_2$        | "pops" with a lighted splint                                                     |
| oxygen, $\text{O}_2$          | relights a glowing splint                                                        |
| sulfur dioxide, $\text{SO}_2$ | turns aqueous acidified potassium manganate(VII) from purple to colourless       |

**(d) Colour of halogens**

| <i>halogen</i>          | <i>colour of element</i>   | <i>colour in aqueous solution</i> | <i>colour in hexane</i> |
|-------------------------|----------------------------|-----------------------------------|-------------------------|
| chlorine, $\text{Cl}_2$ | greenish yellow gas        | pale yellow                       | pale yellow             |
| bromine, $\text{Br}_2$  | reddish brown gas / liquid | orange                            | orange-red              |
| iodine, $\text{I}_2$    | black solid / purple gas   | brown                             | purple                  |

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